

MD. MAJIDUL ISLAM

AI & ML · GENAI · AGENTIC AI

Dhaka, Bangladesh

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“Building intelligent systems that turn data into decisions and automation into advantage”

Summary

AI/ML engineer and architect with end-to-end ownership of production LLM, RAG and multi-agent systems – research, design, deployment and monitoring. Architect and shipper of live AI platforms: **Empathika** (healthcare – medication, shift and care management) and a **FinTech GraphRAG chatbot** (financial fraud and corporate intelligence over Neo4j). Published and maintain two open-source Python packages on PyPI – **personakit** and **spyy** – with full CI/CD and Trusted Publishing; IEEE-published author on banking cybersecurity (2024); active mentor in the AI/ML practitioner community and PMI member.

Technical Skills

LLM SYSTEMS & MULTI-AGENT ARCHITECTURE

- **Frameworks:** LangChain, LangGraph, CrewAI, AutoGen, LlamaIndex
- **RAG & Vector Stores:** retrieval design, embedding analysis, ChromaDB, Pinecone
- **Providers & Serving:** LiteLLM (100+ providers), OpenAI, Anthropic, DeepSeek, Groq, Ollama, vLLM
- **Agent Safety & Ops:** static AST prompt discovery, deterministic checkers, union/override hybrid verdicts, self-correcting loops, Phoenix, Langfuse, OpenTelemetry

EDGE AI & GPU-ACCELERATED COMPUTING

- NVIDIA Jetson, CUDA, TensorRT, ONNX Runtime, FP16/INT8 quantization, PyTorch, TensorFlow, low-latency inference

ML INFRASTRUCTURE & MLOPS

- FastAPI, Streamlit, MLflow, Docker, GitHub Actions, Jenkins, PyPI Trusted Publishing, AWS, GCP, Azure, MongoDB, Neo4j

DATA & CORE ML

- Pandas, NumPy, PySpark, Scikit-Learn

Work Experience

TulipTech Limited

Dhaka, Bangladesh

AI/ML ENGINEER

Aug. 2025 - Present

- **Built and optimized Edge AI pipelines** on NVIDIA Jetson platforms for low latency inference using TensorRT, CUDA acceleration and FP16/INT8 optimizations
- **Developed multimodal AI systems** combining vision, text and document understanding including OCR pipelines for structured and unstructured data
- **Implemented Retrieval Augmented Generation (RAG)** using LangChain and CrewAI with vector databases for reliable knowledge grounding
- **Designed and orchestrated multi-agent AI systems** using CrewAI and LangGraph for task decomposition, tool usage and collaborative workflows
- **Fine-tuned and evaluated large language models** on domain-specific datasets using in-house GPU clusters, optimizing accuracy
- **Implemented LLM observability and hallucination analysis** using Phoenix and Langfuse for prompt tracing, evaluation, latency and cost monitoring

Synova Information Systems Limited

Dhaka, Bangladesh

AI PROGRAMMER DATA

Mar. 2025 - Aug. 2025

- **Data Driven AI Solutions:** Built and optimized data driven AI solution with backend
- **AI Backend Development:** Create and Implement FastAPI backend for AI tools
- **Agentic AI Solution:** Designed multi-agent AI systems to perform autonomous reasoning and decision-making tasks
- **Big Data Processing:** Processed large datasets using tools like Pandas, PySpark for real time insights
- **MLOps Implementation:** Set up automated pipelines for model training, testing, deployment and monitoring using tools like MLflow, LlamaIndex
- **LLM Oriented Automation:** Automated reasoning, task execution, and content generation using large language models
- **Multi Agent Framework Integration:** Leveraged frameworks like LangChain, CrewAI and AutoGen to orchestrate collaboration between AI agents

TulipTech Limited

SOFTWARE ENGINEER(AI)

Dhaka, Bangladesh

Oct. 2022 - Mar. 2025

- **Designed and Deployed AI Solutions** using Python and FastAPI following MLOps best practices for scalable and maintainable machine learning workflows
- **Implemented End to End ML Architecture** for training, testing, deployment and monitoring, ensuring robust model life cycle management
- **Led Cloud & DevOps Integration** with Microsoft Azure to provision, monitor and optimize cloud-based AI infrastructure
- **Engineered FAST APIs and Data Architectures** maintaining performant, secure APIs and managing document based databases (MongoDB)
- **Contributed to Project Management** supporting agile delivery through cross-team coordination and CI/CD pipeline maintenance
- **Multi Agent Framework Integration:** Leveraged frameworks like LangChain, CrewAI and AutoGen to orchestrate collaboration between AI agents

Bangladesh Navy

OFFICER CADET

Bangladesh Naval Academy,

Bangladesh

Dec. 2016 - Mar. 2018

- **Leadership & Discipline:** Completed intensive military training, strengthening leadership, teamwork and decision-making under pressure
- **Maritime & Tactical Training:** Acquired hands on experience in naval operations, navigation and tactical combat strategies
- **Strategic Planning & Crisis Management:** Trained to operate effectively in high-stress environments, making quick and strategic decisions
- **Technical & Weaponry Knowledge:** Gained foundational knowledge in naval engineering, weapon systems and maritime technologies
- **Ethics & Professionalism:** Maintained strong discipline, integrity and commitment to service throughout military service

Projects

Author | PyPI | GitHub

MIT License

SPYV — PROMPT-SECURITY TESTING FOR AI ENGINEERS (OPEN SOURCE)

2026 – Present

- **Author and maintainer** of an open-source, framework-agnostic tool that discovers every LLM prompt in a codebase (CrewAI, LangChain, OpenAI) via AST – no code execution – then audits, red-teams and guards them at runtime.
- **Deterministic checkers** for secrets, PII and prompt-leaks (no LLM needed) fused with a hardened LLM judge via union/override hybrid verdicts; self red-team tests prove verdicts cannot be flipped.
- **Runtime @guard** observes real agent output as ground truth and blocks leaks; bring-your-own-model (OpenAI, Anthropic, Gemini, vLLM/Ollama). Live on PyPI with full CI/CD.

Author | PyPI | GitHub

MIT License

PERSONAKIT — DECLARATIVE AGENT BUILDER FOR LLMs (OPEN SOURCE)

2026 – Present

- **Author and maintainer** of an open-source Python library that builds role-based LLM agents declaratively (persona, frameworks, probes, red flags as data) – domain-agnostic (code review, fintech AML, support triage, clinical triage).
- **Cross-provider runtime** with streaming, OpenTelemetry hooks, token-cost tracking and a multi-turn tool loop – OpenAI, Anthropic, plus 100+ providers via LiteLLM.
- **Red flags** as a first-class primitive (trigger → severity → action → citation, regex AND semantic LLM); 11 PyPI releases, 93 tests, `mypy --strict` clean, full CI/CD.

Live in Production | Multi-Agent RAG | Medication, Shift & Care

Healthcare AI

EMPATHIKA — AI PLATFORM FOR HEALTHCARE & RESEARCH

2025 – Present

- **Currently live in production** – healthcare AI platform serving real users across medication, shift and care management.
- **Medication:** OCR + AI-booster extraction pipeline delivering faster structured output at 97% accuracy.
- **Care Management:** multi-agent RAG over clinical records + external medical knowledge for grounded, citation-backed care planning – powered by personakit in production.

Live | Multi-Agent GraphRAG | Fraud & Corporate Intelligence

FinTech | Demo:

agentic-corporate-forensics

FINTECH CHATBOT — FINANCIAL GRAPH RAG (NEO4J)

2026 – Present

- **Live** financial GraphRAG chatbot – multi-agent system (Orchestrator, Translator, Auditor, Forensic Analyst) turning natural-language questions into Cypher over a Neo4j graph to trace transaction flows, beneficial ownership and conflicts of interest.
- Detects money-laundering paths, shell / nominee structures and circular payment loops with a self-correcting Cypher loop; public demo open-sourced.

Sales forecasting (Big Data)

Dhaka, Bangladesh

AI SOLUTION – SALES FORECASTING

2025

- Developed a machine learning based sales forecasting system leveraging historical sales data, seasonal trends and external factors.
- Enabled data driven decision making for marketing, inventory and supply chain teams through predictive analytics dashboards.

Publications

Multi Factor Authentication Kerberos

IEEE SEM Vol -12, 2024

INTERNET BANKING CYBER SECURITY IN BANGLADESH

Jan. 2024

- Multi factor authentication solutions for Bangladeshi Banks

Education

Southeast University, Bangladesh BSC. IN COMPUTER SCIENCE AND ENGINEERING CGPA: 3.66 / 4.00	<i>Dhaka, Bangladesh</i> <i>Oct. 2018 – Oct. 2022</i>
Adamjee Cantonment College HIGHER SECONDARY CERTIFICATE GPA: 5.00 / 5.00	<i>Dhaka, Bangladesh</i> <i>Aug. 2016</i>
Muslim Modern Academy SECONDARY SCHOOL CERTIFICATE GPA: 5.00 / 5.00	<i>Dhaka, Bangladesh</i> <i>May. 2014</i>

Memberships, Training and Certifications

PROFESSIONAL MEMBERSHIPS

2026 – Present	Member , Project Management Institute (PMI)	<i>Global</i>
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PROFESSIONAL CERTIFICATIONS

2025	Oracle Certified Foundation Associate , Certified AI Foundation	<i>Oracle University</i>
2024	Agile Project Management Practitioner , Project Management	<i>HP LIFE</i>
2023	Admin Official Certification , Project Management	<i>monday.com</i>
2022	Cyber Security (Ethical Hacking) , Arena Web Security	<i>Dhaka, Bangladesh</i>

TRAINING

2023	Industrial Data Science and Machine Learning with AI , HR Ventures	<i>Dhaka, Bangladesh</i>
2021	Ethical Cracking and Hacking , SkillNation Bangladesh	<i>Dhaka, Bangladesh</i>

Extracurricular Activity

AI/ML Professional Community LinkedIn Group MENTOR - Mentor practitioners on production AI/ML topics: LLM systems, RAG pipelines, multi-agent architectures, MLOps, edge deployment and career growth. - Share open-source work and engineering writeups with the community, supporting members building real-world AI products.	<i>Online Global</i> <i>2025 – Present</i>
96.4 Spice Fm RADIO JOCKEY - Created content, wrote scripts and conducted live interviews with artists, entrepreneurs and public figures to drive audience engagement. - Collaborated with producers and broadcast team to ensure smooth broadcasting, while maintaining a lively and energetic on air presence	<i>Dhaka, Bangladesh</i> <i>Oct. 2020 - Oct. 2022</i>